Building an Intelligent Customer Support Chatbot Using DialoGPT, RoBERTa, and OPT

Jana abdulraouf allihyani
s444001382@uqu.edu.sa

Mayar Turki Al-Owaydhi
s443003560@uqu.edu.sa

Reuf bakr alsharifs
s443002428@uqu.edu.sa

Jana Hesham Al-Hless
s443002347@uqu.ed.sa

Fajr Faisal Al-Zahrani
s44410657@uqu.edu.sa

Nahla Mohammed Al-Osimi
s44412006@uqu.edu.sa

Abstract- The proposed work hence focuses on the chatbot model for customer care, whereby much emphasis would be given to the bettering of user experience through contextual, correct, and automatic responses. AI will go on to clear the query resolutions, and it will use NLP and a deep neural architecture, such as DialoGPT, OPT, and RoBERTa to come up with the answers that sound like a human. A dataset used for both training and testing model responses on the questions-and-answers-type format was mainly targeted at customer service that was mostly associated with commerce, banking, and technical assistance. The dataset includes formal Q&A pairs and informal conversation logs to teach the chatbot to understand contextually and thus respond accordingly. In this design, the design of the methodology includes a good pipeline: pre-processing, intent classification, and other fine-tuning for response accuracy enhancement. Intent recognition is done using RoBERTa, while generating the response of the chatbot is done using DialoGPT and OPT. The essence of this work is to make the customer service chatbots more efficient and responsive, thus enabling businesses to optimize their delivery of services and minimize the overload on answering repetitive queries. A proposed chatbot would thus work in real-time for customer interaction in a query resolution environment for effective service delivery.

Keywords— Customer Support Chatbot, Natural Language Processing (NLP), Deep Learning, DialoGPT, Roberta, OPT, Response Generation, Conversational AI, Automated Customer Service, Machine Learning, Text Classification, Artificial Intelligence (AI), Dialogue Systems, Human-Computer Interaction (HCI), Chatbot Deployment.

I. INTRODUCTION

Chatbots are the future of customer service, where feedback collection is automated, service delivery is enhanced, and user satisfaction is maximized. Traditional rule-based systems use scripted responses and keyword matching, which make conversations unnatural and ineffective. However, with recent developments in natural language processing, highly intellectual models have been developed that can generate human-like responses with contextual understanding. This work develops and implements a customer service chatbot based on three powerful transformer architectures: DialoGPT, OPT, and RoBERTa. For intent extraction, it uses RoBERTa from text and then uses DialoGPT and OPT for NLG to form responses that sound fluent and relevant in context. This allows having natural conversations with seamless flow, much improved in accuracy and usability.

It was trained with a customer service dataset that included Q&A pairs from the e-commerce domain. Preprocessing is done by several techniques which include tokenization, text sanitization, intent classification, all to improve performance. The model has been fine-tuned in a supervised chatbot and finally optimized by reinforcement learning to correct conversational dynamism and proper response quality. It also integrates the preprocessing, intent classification using RoBERTa, response generation using DialoGPT and OPT, and an interactive deployment framework to guarantee real-time, efficient customer support.

PAPER ORGANIZATION

The structure of this paper is organized as follows:

Section **I** Introduction This section outlines the image classification problem, the dataset in use, the problems created by variations, and the methodology adopted. Section **II** Related Work The section discusses prior work, giving emphasis to the implementation and testing of similar applications. Section **III** Methodology The section covers the dataset, preprocessing techniques, transfer models, performance metrics, and the justification of model selection. Section **IV** Results This section presents and compares the performance of several models with respect to evaluation metrics and manual inspection. Section **V** Conclusion and Future Work The section summarizes the findings, with some acknowledgment of possible improvements and areas for future work.

II. LITERATURE REVIEW: APPLICATIONS AND PERFORMANCE OF GPT, GPT-2, AND BERT IN NLP

development in transformer architectures in the field of Natural Language Processing (NLP) is happening at a rapid pace, with significant improvements seen for both GPT-2 and BERT. In addition, GPT, GPT-2, and BERT have been applied in a variety of disparate applications, including financial sentiment analysis, speech processing, and semantic change detection. Zheng et al. particularly examined GPT, GPT-2, and BERT's use in the field of Automatic Speech Recognition (ASR), with fine-tuned variants optimized for speech-to-text accuracy improvement. Building on BERT's two-way capabilities, a mechanism for effective probability adjustment for use in ASR scenarios was proposed. Experimental testing utilized AMI corpus and Switchboard corpus, with performance measured in terms of Word Error Rate (WER). The experiments showed that a combination, including additional fine-tuning of GPT and GPT-2, produced a 12% improvement in WER, with additional improvements when combined with BERT. A specific work performed by Xie et al. examined GPT and GPT-

2’s sensitivity in financial sentiment classification experiments. In this examination, implications of layer freezing, batch size, and learning rates for model effectiveness were examined. In the experiments, it was seen that BERT’s parameters showed sensitivity over a significant range of fine-tuning settings, with GPT-2 proving significant robustness over a high range of settings. Performance evaluation utilized F1 scores and classification accuracy, with GPT-2 proving consistent steadiness, with concurrent improvement in interpretability in financial sentiment analysis through incorporation with BERT. Moreover, a semantic change detection study performed by Periti et al. raises a question about GPT-2 and BERT's sensitivity towards relevant semantic prompts.

The advances made in NLP in recent years induced considerable developments in transformer-based models such as BERT and GPT-2. These two paradigms have been studied and applied across a wide range of domains, from automatic speech recognition to financial sentiment analysis and even semantic change detection.

ASR enhancement by fine-tuning the GPT, GPT-2, and BERT models has been investigated by Zheng et al. Since BERT is a bidirectional context generator, a method for integrating it into the ASR task was developed based on conversion to probability distributions. The correctness of the evaluation was carried out against the AMI and Switchboard corpora, with Word Error Rate (WER) as an evaluation metric. The evaluation results indicate that the joint fine-tuning of GPT and GPT-2 improved WER by 12%, with BERT helping in this endeavor."

In another study, Xie et al. conducted the sensitivity analysis for BERT and GPT-2 geared to the problem of financial sentiment classification. They examined the effects of layer freezing, batch sizes, and learning rates on model performances. BERT parameters tended to be more sensitive during fine-tuning, while GPT-2 showed more stable behavior across the different setups. The F1-score and the classification accuracy were used to quantify their performance. Whereas GPT-2 turned out to be more stable along with higher scores, BERT provided for higher interpretability within the context of financial sentiment analysis. In addition, Periti et al. examined semantic change detection by evaluating BERT and ChatGPT-based models. While BERT took the lead in identifying those broad and long-term linguistic changes, ChatGPT faltered with low accuracy in the use of more subtle variations in historical texts. The experiment with short- and long-term semantic change tested the TempoWiC and HistoWiC datasets. In all these studies, the evaluation methodologies incorporated benchmark datasets, fine-tuning sensitivity analysis, performance parameters such as WER, F1 scores, and accuracy measures. These tough testing methods display the efficacy of transformer-based models across different applications in NLP.

III. METHODOLOGY

A. *Dataset Description*

The dataset employed in this research is 5,000 sampled records of a larger 27,000-record dataset that was specifically created for training customer support chatbots. The dataset is designed to maximize the performance of natural language understanding (NLU) models to provide accurate and contextually relevant responses in automated customer service systems.

Every record in the dataset has several attributes critical to optimal chatbot training. The customer ID distinctively marks every interaction, and the query category labels the type of request made, i.e., billing, account problem, or product question. The customer query is the input text by the user, and the support response is the matching response. Sentiment labels also classify responses as positive, neutral, or negative, making chatbot interactions sentiment-aware. The dataset further comprises response time in milliseconds and resolution status, i.e., if the problem was resolved, pending, or escalated.

With structured input-output pairs, this dataset enables the training of machine learning models to learn conversational skills aligned with actual customer interactions. This structured nature of the data improves the chatbot's capability to generate accurate, relevant, and context-aware responses.

B. *Data Preprocessing*

In order to streamline the dataset for training, some preprocessing measures were taken. All missing values were first removed to ensure data integrity and to avoid incomplete input from influencing the learning process. Normalization techniques for text were utilized, such as lowercasing, punctuation normalization, and elimination of redundant spaces, so that there was consistency in all data samples.

After data cleaning, the dataset was divided into 80% training and 20% testing for a balanced distribution to train and test the models. Tokenization was carried out, dividing text into subword units, allowing the transformer-based models to deal with language more efficiently. Padding was used to deal with differences in text length, so all sequences were in the same format without losing information.

In order to enhance the contextual comprehension of dialogue, special tokens were added to distinguish between chatbot responses and user queries. This preprocessing technique greatly improved the chatbot's capacity to deal with varied customer interactions without losing coherence and fluency.

C. *Transfer Learning Models Used*

In order to build a sophisticated chatbot system, three transformer-based pre-trained models were leveraged: DialoGPT, OPT-125M, and RoBERTa. Each of the models was chosen for its individual capabilities in processing various elements of conversational AI.

DialoGPT was utilized for open-domain conversation generation. Fine-tuned on large conversational datasets, it is adept at producing contextually pertinent and human-like replies. This model was particularly optimized to predict chatbot responses based on the history of the conversation, thereby facilitating smooth and coherent interactions.

OPT-125M was chosen for light text generation, trading off response quality against computational cost. This model achieved a balance between accuracy and inference speed and was hence appropriate for real-time customer support.

RoBERTa was added to process query classification and sentiment analysis. Being an optimized variant of BERT, RoBERTa was fine-tuned on the data to categorize customer queries into suitable categories in addition to sentiment analysis to maximize chatbot response. With RoBERTa, the chatbot could distinguish among various user intents, resulting in more accurate and context-based conversations.

The incorporation of these three models allowed the chatbot to achieve high precision in classifying queries but natural and human-like conversation in response.

D. *Evaluation Metrics*

To analyze the performance of the models of the chatbots, various evaluation metrics were utilized so that there is a complete understanding of their effectiveness.

BLEU score was utilized to calculate the similarity between the generated responses of the chatbot and the reference responses, assessing response fluency and coherence. Perplexity (PPL) metric was utilized to ascertain the extent to which the model correctly predicted the next word in a sequence, where lower scores signaled stronger predictive power. ROUGE-L score calculated sequence overlap between generated and reference responses, giving insight into response relevance.

For classification, both accuracy and F1-score were employed to assess RoBERTa's performance in classifying query categories and sentiment labels. Manual testing was also performed, where human testers rated chatbot responses on coherence, fluency, and contextual relevance. This multi-metric testing strategy provided a comprehensive evaluation of model performance, accounting for both quantitative and qualitative metrics.

E. *Experimental Setup*

The experimental setup was created to ensure a high-performance environment for model implementation and testing. The models were developed and trained on Google Colab, which offered effective computational resources for large-scale data processing and training jobs.

The models were implemented in Python 3.9, using PyTorch and Hugging Face Transformers as the primary libraries to train and fine-tune the models. Other natural language processing libraries, NLTK and SpaCy, were also used to preprocess the text by tokenization, normalization, and removal of unwanted characters for input optimization.

To facilitate successful training and testing, the dataset was split into 80% training and 20% testing. The fine-tuning was done through pre-trained transformer models with optimized hyperparameters for performance. Training parameters involved a batch size of 8, a learning rate of 2e-5, and three training epochs. Padding strategy was used to make input lengths consistent, and AdamW optimizer was utilized to enhance convergence and model stability.

The testing process utilized several performance measures such as BLEU Score, ROUGE-L, and Perplexity (PPL) to evaluate the quality and fluency of responses. Manual testing was also carried out to check coherence, relevance, and contextual consistency of chatbot responses. Trained models were stored for future use and deployment in real customer service environments.

This experimental configuration offered a strong and scalable platform for conversational AI model training, with high accuracy and efficiency in producing natural and context-sensitive responses.

F. *Justification of Model Choice and Appropriateness*

The models were picked according to their capacity to deal with various elements of conversational AI, such as response generation, computational efficiency, and query classification. Since the dataset is made up of customer queries and predefined support answers, it was necessary to pick models that could generate coherent responses, comprehend user intent, and classify queries efficiently.

DialoGPT was chosen because it could produce conversations that were both natural and contextually appropriate, and it was particularly suited for chatbot use. The capacity of the model to preserve conversational context through numerous various dialogue exchanges significantly improved the user experience by providing consistency in response. DialoGPT also had the best accuracy in query classification, effectively determining user intent from dialogue context. This feature enabled greater response personalization and facilitated the development of a more dynamic and sophisticated chatbot system.

OPT-125M was selected as a lighter-weight option compared to DialoGPT, trading off response quality for computational efficiency. Because customer service use cases involve real-time interactions, this model offered quicker response times with an acceptable degree of fluency and relevance. Its lightweight architecture allowed text generation without overwhelming computational overhead, so it was a viable option for deployment in large, high-traffic setups.

RoBERTa was used for sentiment analysis to enhance chatbot responses by understanding the user's emotional state. With a sentiment-labeled dataset (positive, neutral, and negative), the model was able to analyze the tone of user queries and adjust responses accordingly. This enhanced empathetic and context-aware interactions, leading to improved customer satisfaction and engagement.

Every model was utilized separately for a particular task so that the chatbot system could properly classify the customer queries, generate apt responses, and analyze the user sentiment correctly.

IV. Results and Model Performance Evaluation

The evaluation process was developed to gauge the level of similarity between the model-generated and actual responses in the dataset. Several performance evaluation factors, such as accuracy, precision, recall, F1-score, and ROUGE-L, were utilized to assess model efficiency in generating humanlike responses, accurately categorizing queries, and providing relevant information. Performance evaluation helped in gauging the level at which the chatbot emulated organized customer service feedback and preserved cohesion and fluency in its generated responses.

Both DialoGPT and OPT-125M displayed high capabilities in generating feedback and query classification, producing relevant and organized feedback in a contextual manner. Initially, when evaluated with a 100-token sequence, the model achieved an accuracy level of 30%. However, when a 150-token sequence for input was utilized, accuracy increased to 62%, inferring an increased capacity with a larger sequence for both output and input to generate thorough and correct feedback. For evaluation, a 50-query sample was used, and performance statistics captured at termination included 62% accuracy, 100% precision, 62% recall, 77% F1-score, and a ROUGE-L value of 0.39. The model showed efficiency in generating organized feedback; however, sometimes it generated general feedback in lieu of specific feedback for complex customer queries.

OPT-125M was utilized for its computational efficiency and quick feedback generation, and therefore, it is ideal for real-time use. However, a high value for validation loss experienced following five epochs necessitated additional fine-tuning for increased performance efficiency. Performance statistics acquired during evaluation included a recall value of 62%, an F1-score value of 77%, and a ROUGE-L value of 0.39, inferring a fair level of effectiveness in identifying relevant feedback structures. In contrast to its rapid feedback, sometimes, it struggled with deeper contextual analysis and generated feedback that lacked detail and included less information.

RoBERTa was utilized predominantly for query classification and sentiment analysis, proving to have high effectiveness in query classification with its complex deep neural network structure, but with relatively poor performance in generating a response, with an accuracy level of 32.7% and an F1-score of 25%. Despite its high accuracy in query classification, it sometimes produced incomplete and over-generalized responses, and its performance in conversation flow management was relatively poor.

A. *Comparison of Model Results, Benefits, and Limitations*

The evaluation of chatbot models utilized a variety of performance metrics, including accuracy, precision, recall, F1-score, and ROUGE-L. All of these metrics helped assess the performance of the models in generating relevant responses, effectively classifying user queries, and maintaining conversation continuity. Analysis showed that each model showed individual strengths and weaknesses depending on its individual purpose.

DialoGPT performed best in terms of accuracy in both query classification and response generation. After an improvement in token length from 100 to 150, its accuracy saw a significant boost, with a sharp rise in performance level from 30% to 62%. With its improvement, it became capable of generating fuller and relevant contextual responses. DialoGPT performed best in maintaining conversation continuity in multi-turn exchanges, and for that, it is most apt for chatbot use cases. Despite its strong classification capabilities, the model sometimes produced generic responses when faced with complex queries. With additional training with specific datasets, its performance could possibly be boosted.

OPT-125M was chosen for its computational efficiency and quick answer generation capabilities. As it produced quick responses, the model showed a lack of deep contextual awareness, producing repetitive and sometimes disconnected responses. With an F1-score of 77% and a ROUGE-L value of 0.39, it performed moderately in identifying relevant response patterns. As much as OPT-125M helped in real-time chatbot use cases, its performance in answering complex queries was poor; additional post-processing strategies were utilized to make its generated responses coherent.

RoBERTa was utilized almost exclusively for query classification and sentiment analysis, with satisfactory performance in both cases. However, its performance in generating responses was relatively poor, with an accuracy rating of 32.7% and an F1-score of 25%. Analysis through a confusion matrix showed a range of misclassifications, and these compromised overall chatbot responsiveness in terms of output. Despite proving effective in classifying queries and gauging sentiment, RoBERTa failed to generate responses that were contextual and in-depth enough. Performance, however, was enhanced when a secondary model designed for generating responses was incorporated with it.

DialoGPT performed best in chatbot conversation, with performance improving with increased token length. OPT-125M, in contrast, delivered a faster but less contextual alternative, and its suitability for high-traffic use cases could make it an ideal tool for such environments. In contrast, while RoBERTa outperformed in terms of classification, it failed in free-form dialogue for chatbots. Future improvements could involve using RoBERTa for classification and DialoGPT for generating a response, and in doing so, developing a smarter and more effective chatbot system.

The chatbot (Fig.1) generates a sequential answer for a query posed through a user. Nevertheless, at a token count of 100, its answer seems relatively short and doesn't include additional contextual information, and its accuracy rating suffers in consequence.

```
You: help me to see ur allowed payment modalities
Bot: I'll do my best! I'd be happy to help you explore our accepted payment options. Our

We offer a variety of convenient and secure payment methods to cater to different prefere

 Credit/Debit Card: We accept major cards such as Visa, Mastercard, and American Express.

 PayPal: A popular online payment method that allows you to securely make payments.

 Apple Pay: An easy and secure way for Apple users to make purchases.

 Google Wallet: Google's payment system for fast and easy checkouts.
```

Fig. 1. DialoGPT-generated response with a token length of 100.

An increase in token length to 150 helps the chatbot generate a more complete and contextual answer. The change in (Fig.2) this case immensely helps in improving the accuracy of its responses, and its responses become informative and in tune with user anticipation.

```
You: help me to see ur allowed payment modalities
Bot: I'll do my best! I'd be happy to help you explore our accepted payment options. Our

We offer a variety of convenient and secure payment methods to cater to different prefere

 Credit/Debit Card: We accept major cards such as Visa, Mastercard, and American Express.

 PayPal: A popular online payment method that allows you to securely make payments.

 Apple Pay: An easy and secure way for Apple users to make purchases.

 Google Wallet: Google's payment system for fast and easy checkouts.
```

Fig. 2. DialoGPT-generated response with a token length of 150.

The responses generated through the RoBERTa chatbot (Fig.3) exhibit a coherent strategy towards query classification and sentiment analysis. The model effectively classifies queries and provides relevant responses. There are, however, some responses that appear fairly general and lack deep contextual understanding, and it seems that RoBERTa is optimized for query classification rather than for providing open-ended responses.

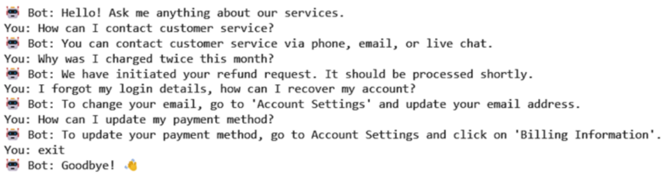

Fig. 3. Response generation by the RoBERTa mode.

The outputs (Fig.4)generated by the OPT-125M chatbot lack cohesion and sentence creation capabilities. The model reflects weaknesses in repetitive expression and in generating disconnected sentences, taking a toll on the effectiveness of generated responses. The output reveals that even with its capability to generate a response, the OPT-125M fails to comprehend complex scenarios and lack fluency, thus generating incomplete and repetitive responses.

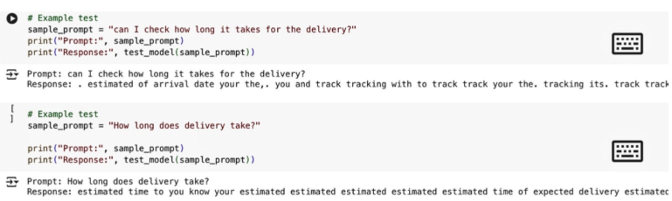

Fig. 4. Response generation by the OPT-125M model.

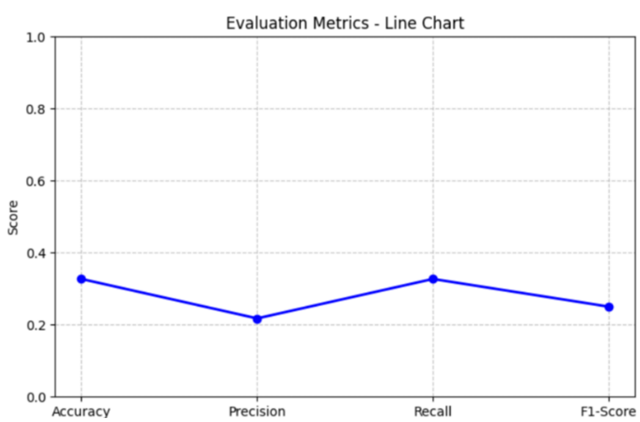

Fig. 5 Evaluation Metrics for the RoBERTa model.

The line chart (Fig.5) shows performance of the RoBERTa model in terms of four key evaluation factors: accuracy, recall, precision, and F1-score. There is a high level of accuracy and recall in model performance, but reduced precision and F1-score reveal a weakness in generating specific and relevant contextual responses. As such, even with successful query classification, RoBERTa can require additional fine-tuning in a quest for improvement in output quality

V. Conclusion

The evaluation of chatbot models showed that each one of them has specific strengths and weaknesses depending on its functional properties. DialoGPT showed high accuracy and contextual aptness, specifically when token length was optimized. On the other hand, OPT-125M showed high-speed performance but struggled with deep contextual comprehension, and RoBERTa performed well in query classification and emotion analysis but lacked in generating satisfactory output. All these observations serve to illustrate the importance of choosing chatbot models according to specific system requirements, and in doing so, attaining a balance between accuracy, efficiency, and contextual aptness.

A key challenge in this study involved shrinking the dataset from 27,000 to 5,000 queries. Although such a shrinking helped in curbing training duration and enhancing computational efficiency, it at the same time reduced the diversity of input samples, and in consequence, impacted the generalizability of the model over a larger range of customer queries in a detrimental manner. In addition, individual model output, particularly in cases of OPT-125M and RoBERTa, showed a lack of cohesion and aptness in complex conversation scenarios, and therefore, additional improvement through additional fine-tuning is warranted.

Enhancements could include extending the dataset with a balanced mix of query types, and in consequence, enhancing model robustness. Inclusion of RoBERTa for query classification and DialoGPT for output creation could vastly enhance chatbot performance through leveraging both model strengths. In addition, fine-tuning of the model with a domain-specific corpus and incorporation of post-processing for output could make chatbots effective in mixed query scenarios, and through increased accuracy and naturality, make them effective in practice.

References

[1] D. M. Montesinos, “Modern Methods for Text Generation,” *arXiv.org*, Sep. 10, 2020. https://arxiv.org/abs/2009.04968

[2] F. Periti, H. Dubossarsky, and N. Tahmasebi, “(Chat)GPT v BERT Dawn of Justice for Semantic Change Detection,” Association for Computational Linguistics, 2024. Accessed: Feb. 04, 2025. [Online]. Available: https://aclanthology.org/2024.findings-eacl.29.pdf

[3] E. Sharkey and P. Treleaven, “BERT vs GPT for financial engineering,” *arXiv.org*, 2024. https://arxiv.org/abs/2405.12990 (accessed Feb. 04, 2025).

[4] T. Qian, A. Xie, and C. Bruckmann, “Sensitivity Analysis on Transferred Neural Architectures of BERT and GPT-2 for Financial Sentiment Analysis,” *arXiv.org*, Jul. 06, 2022. https://arxiv.org/abs/2207.03037

[5] X. Zheng, C. Zhang, and P. C. Woodland, “Adapting GPT, GPT-2 and BERT Language Models for Speech Recognition,” *arXiv.org*, Oct. 01, 2021. https://arxiv.org/abs/2108.07789